

Microeconomics

Lecture 13: Competitive Equilibrium

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What We Know

Demand side:

- The consumer maximizes utility: $MRS = p_x/p_y$
- The market demand curve aggregates individual demands (horizontally)
- Demand is decreasing in price; shifted by income, prices of related goods, preferences

Supply side:

- The price-taking firm produces where $P = MC$ (with MC increasing)
- The market supply curve aggregates individual supplies (horizontally)
- Supply is increasing in price; shifted by input costs

Now we must put them together!

Where demand and supply meet, the **equilibrium price and quantity** are determined. We analyze what happens in the short run and in the long run, and how the equilibrium responds to shocks.

Lecture Objectives

1. **Define** competitive equilibrium and the price adjustment mechanism
2. **Compute** the equilibrium price and quantity by solving $Q^D = Q^S$
3. **Analyze** excess demand and excess supply graphically
4. **Apply** comparative statics to demand and supply shocks
5. **Describe** the long-run adjustment mechanism
6. **Determine** price, quantity and number of firms in the LR equilibrium
7. **Classify** constant-, increasing- and decreasing-cost industries
8. **Solve** complete exercises on SR and LR equilibrium

Focus

Equilibrium is the heart of the competitive model. The SR tells us where we are “now”; the LR tells us where we will end up given enough time.

Definition of Competitive Equilibrium

Market equilibrium

A **competitive equilibrium** is a pair (P^*, Q^*) such that:

$$Q^D(P^*) = Q^S(P^*)$$

The market “clears”: quantity demanded equals quantity supplied. There is neither excess demand nor excess supply.

How to Find It

Solve the system

$$\begin{cases} Q^D = Q^D(P) \\ Q^S = Q^S(P) \\ Q^D = Q^S \end{cases}$$

Technically: substitute the third equation into the first and set it equal to the second, obtaining an equation in P alone.

Example

$$Q^D = 120 - 4P \quad (\text{inverse demand: } P_D = 30 - Q/4)$$

$$Q^S = 20P \quad (\text{inverse supply: } P_S = Q/20)$$

Computing the Equilibrium

Set $Q^D = Q^S$:

$$120 - 4P = 20P \Rightarrow 120 = 24P \Rightarrow \boxed{P^* = 5}$$

$$Q^* = Q^D(5) = 120 - 20 = 100 \quad (\text{or } Q^* = Q^S(5) = 100) \quad \checkmark$$

Check: $Q^D(5) = 100 = Q^S(5) \checkmark$. The market clears: every consumer who wants to buy at price P^* finds a seller, and vice versa.

Next step. The equilibrium is associated with a measurable *gain from trade*: consumer and producer **surplus**. We introduce them in the next section.

The Adjustment Mechanism

How do we get to equilibrium? The price adjusts automatically.

If $P > P^*$: excess supply

- Sellers cannot sell everything $\Rightarrow$ they lower the price
- P falls toward P^*

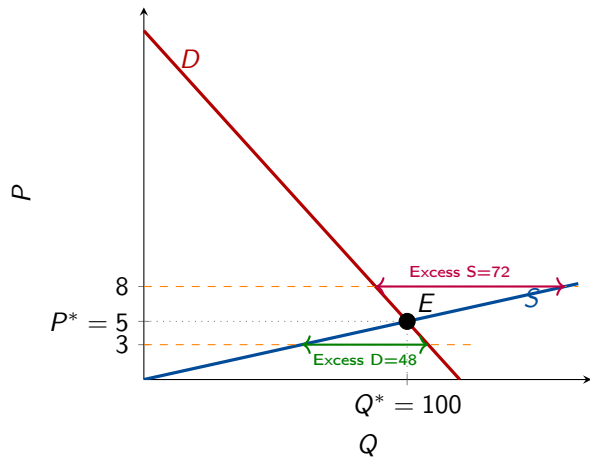
If $P < P^*$: excess demand

- Buyers cannot buy all they want $\Rightarrow$ they offer higher prices
- P rises toward P^*

At $P = 3$ (below equilibrium): $Q^D = 108$, $Q^S = 60$. Excess demand: 48 units.
Consumers compete with each other $\Rightarrow P$ rises toward 5.

At $P = 8$ (above equilibrium): $Q^D = 88$, $Q^S = 160$. Excess supply: 72 units.
Producers compete with each other $\Rightarrow P$ falls toward 5.

Equilibrium: Graph with Excesses



At $P = 3$:

$$Q^D = 108 > Q^S = 60$$

$\uparrow P$ toward 5

At $P = 8$:

$$Q^D = 88 < Q^S = 160$$

$\downarrow P$ toward 5

At $P^* = 5$:

$$Q^D = Q^S = 100$$

Stable equilibrium

The adjustment mechanism is spontaneous: no agent needs to know the overall state of the market.

Existence, Uniqueness, Stability (1)

When does an equilibrium exist?

- $Q^D(P)$ continuous and decreasing in P ;
- $Q^S(P)$ continuous and increasing in P ;
- the two curves intersect in the positive orthant.

Under these standard conditions, the equilibrium **exists and is unique**.

Existence, Uniqueness, Stability (2)

Stability (Walrasian adjustment). The equilibrium P^* is stable if, after a small perturbation, the readjustment force pushes the price back toward P^* :

$$\left. \frac{d}{dP} [Q^S(P) - Q^D(P)] \right|_{P^*} > 0$$

Other cases:

- *Backward-bending* labor supply curve: at very high wages, the income effect dominates and supply falls, with possible multiple equilibria.
- Giffen goods: demand increasing in price for inferior goods with a dominant income effect.

In the rest of the course we always assume the standard conditions: D decreasing, S increasing, unique and stable equilibrium.

Market Clearing in Practice

The Walrasian mechanism is not just a theoretical abstraction. Several real markets work exactly this way: an *auctioneer* (explicit or implicit) updates the price until $Q^D = Q^S$.

Examples institutionally structured as Walrasian auctions:

- **Wholesale fish markets** (Chioggia, Rialto, Milan): open-outcry *descending* auction. The price starts high and falls until a buyer accepts.
- **Italian Treasury auctions** (BTP, BOT): the Treasury announces the quantity, dealers submit bids, the allotment price clears the market.
- Stock exchange **opening auctions** (Milan, NYSE, LSE): before the open, an electronic system computes the price that maximizes traded volume.
- **Day-ahead electricity market** (MGP, GME): for each hour a price is set that equates hourly demand and supply.

The point

In markets that are *not* institutionally organized (supermarkets, services) market clearing happens more slowly and in a fragmented way: firms observe inventories and adjust prices and output. The long-run outcome is the same.

Consumer and Producer Surplus

The competitive equilibrium is not just a geometric meeting point: it is the configuration that realizes a **gain from trade**, measurable in euros.

Consumer surplus (CS)

Difference between what consumers would be willing to pay (*willingness to pay*, WTP, i.e., the demand curve) and what they actually pay at the market price P^* .

$$CS = \int_0^{Q^*} [P_D(q) - P^*] dq = \text{area between } D \text{ and the line } P^*$$

Producer surplus (PS)

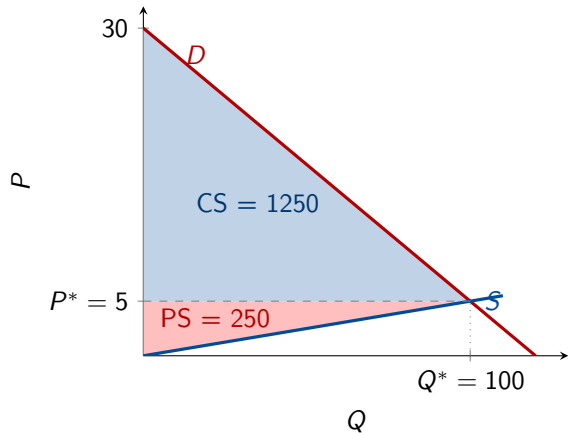
Difference between producers' revenue at price P^* and the marginal costs of production (the supply curve).

$$PS = \int_0^{Q^*} [P^* - P_S(q)] dq = \text{area between the line } P^* \text{ and } S$$

Total surplus: $TS = CS + PS$. For *linear* demand and supply, both areas are triangles.

Computing Surplus: Our Example

Market: $Q^D = 120 - 4P$ (inv. $P_D = 30 - Q/4$), $Q^S = 20P$ (inv. $P_S = Q/20$). **Equilibrium:** $P^* = 5$, $Q^* = 100$.



Consumer surplus:

$$\begin{aligned} CS &= \frac{1}{2} \cdot Q^* \cdot (P_{\max}^D - P^*) \\ &= \frac{1}{2} \cdot 100 \cdot (30 - 5) = 1250 \end{aligned}$$

Producer surplus:

$$\begin{aligned} PS &= \frac{1}{2} \cdot Q^* \cdot P^* \\ &= \frac{1}{2} \cdot 100 \cdot 5 = 250 \end{aligned}$$

Total surplus:

$$TS = 1,250 + 250 = 1500$$

The asymmetry ($CS \gg PS$) reflects the difference in slopes: D is steeper than S , so consumers capture a larger share of the gain from trade.

CS as Willingness to Pay: A Discrete Example

Five consumers each want one beer. Their **WTPs** are:

$$\text{WTP} = \{10, 8, 6, 4, 2\} \text{ euros}$$

The market price is $P = 5$ euros. Who buys?

Consumer	WTP	Buys?	Individual surplus
1	10	yes	$10 - 5 = 5$
2	8	yes	$8 - 5 = 3$
3	6	yes	$6 - 5 = 1$
4	4	no	—
5	2	no	—
Total CS			9 euros

Moving to the continuous case. With many consumers with different WTPs, the sum becomes an integral:

$$\text{CS} = \int_0^{Q^*} [P_D(q) - P^*] dq$$

Geometrically, surplus is the *area* below the demand curve and above the horizontal line $P = P^*$.

PS as Willingness to Sell: A Discrete Example

Five firms can each produce one beer. Their **marginal costs** are:

$$MC = \{1, 2, 3, 4, 5\} \text{ euros}$$

The market price is $P = 4$ euros. Who produces?

Firm	MC	Produces?	Individual surplus
1	1	yes	$4 - 1 = 3$
2	2	yes	$4 - 2 = 2$
3	3	yes	$4 - 3 = 1$
4	4	yes	$4 - 4 = 0$
5	5	no	—
Total PS			6 euros

Moving to the continuous case. $PS = \int_0^{Q^*} [P^* - P_S(q)] dq$. Geometrically: area above the supply curve and below the horizontal line $P = P^*$.

Note. PS is not accounting profit: it is revenue minus variable costs. It coincides with profit only if there are no fixed costs.

The First Welfare Theorem

First Fundamental Theorem of Welfare Economics

In a perfectly competitive market, the competitive equilibrium (P^*, Q^*) maximizes total surplus TS.

Intuition. Every unit traded at P^* satisfies two conditions:

- a consumer with $WTP \geq P^*$ receives it, so CS is non-negative;
- a firm with $MC \leq P^*$ produces it, so PS is non-negative.

All units such that $WTP \geq MC$ are traded; all those with $WTP < MC$ are not traded. **No efficient trade is left unrealized.**

Practical consequence. Any *deviation* from the competitive equilibrium—quantity traded $\neq Q^*$, price $\neq P^*$ —generates a **deadweight loss** (DWL): TS falls below 1,500.

Comparative Statics

Comparative statics

Comparative statics compares two different equilibria—before and after a shock—without describing the adjustment path. It tells us the **direction** and the **size** of the changes.

Procedure:

1. Identify the shock (demand or supply? in which direction?)
2. Compute the new equilibrium
3. Compare P^{**} , Q^{**} with P^* , Q^*

Comparative Statics: Summary Table

Shock	Curve	Direction	ΔP^*	ΔQ^*
Demand increase	D	$\rightarrow$	$\uparrow$	$\uparrow$
Demand decrease	D	$\leftarrow$	$\downarrow$	$\downarrow$
Supply increase	S	$\rightarrow$	$\downarrow$	$\uparrow$
Supply decrease	S	$\leftarrow$	$\uparrow$	$\downarrow$

Demand Shock: Analysis

Scenario: incomes rise. Demand for the (normal) good shifts right: Q^D becomes $Q^{D'} = 140 - 4P$.

New equilibrium:

$$140 - 4P = 20P \Rightarrow 140 = 24P \Rightarrow P^{**} = \frac{35}{6} \approx 5.83$$

$$Q^{**} = 20 \cdot \frac{35}{6} = \frac{350}{3} \approx 116.7$$

Effects: P rises from 5 to 5.83 (+0.83); Q rises from 100 to 116.7 (+16.7).

Real-world example—COVID-19 pandemic and face masks (2020). Demand for face masks rose vertically in March 2020. With supply rigid in the short run, prices shot up (from €0.50 to over €5 per mask). A demand shock with inelastic supply in the SR.

Supply Shock: Analysis

Scenario: the cost of raw materials rises. The supply curve shifts left: Q^S becomes $Q^{S'} = 20P - 80$ (equivalent to a cost increase of €4 per unit).

New equilibrium:

$$120 - 4P = 20P - 80 \Rightarrow 200 = 24P \Rightarrow P^{**} = \frac{25}{3} \approx 8.33$$

$$Q^{**} = 120 - 4 \cdot \frac{25}{3} = 120 - \frac{100}{3} = \frac{260}{3} \approx 86.7$$

Effects: P rises from 5 to 8.33; Q falls from 100 to 86.7.

Real-world example—Russia-Ukraine war and wheat (2022). Russia and Ukraine supply about 30% of world wheat exports. The Russian invasion of February 2022 reduced available supply, pushing the FAO wheat price index up by +60% within a few months. A negative supply shock.

Simultaneous Shocks: Demand and Supply

What happens when both curves shift?

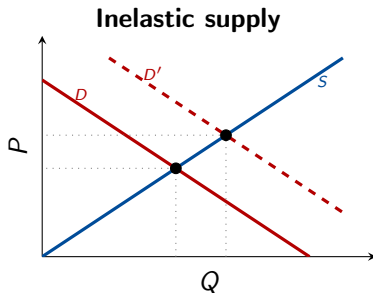
Rule: when a shock shifts only one curve, both ΔP^* and ΔQ^* have a definite sign. When both shift, one of the two effects is ambiguous.

D and S shift...	ΔQ^*	ΔP^*	Ambiguous
Both right ($D \uparrow, S \uparrow$)	$\uparrow$ certain	?	P^*
Both left ($D \downarrow, S \downarrow$)	$\downarrow$ certain	?	P^*
In opposite directions ($D \uparrow, S \downarrow$)	?	$\uparrow$ certain	Q^*
In opposite directions ($D \downarrow, S \uparrow$)	?	$\downarrow$ certain	Q^*

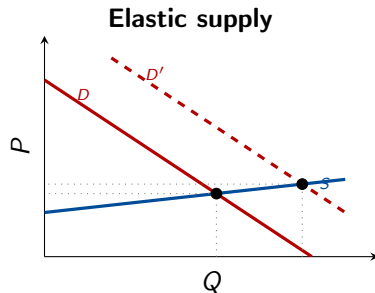
Example—energy crisis and post-COVID recovery. Rising energy costs (S shifts left) and the post-lockdown demand recovery (D shifts right) acted in opposite directions: the certain effect is $\uparrow P^*$, but the sign of ΔQ^* depends on the relative size of the shocks.

Elasticity and the Incidence of Shocks

Empirical observation. The same demand shock has very different effects on P and Q depending on the elasticity of supply.



ΔP large, ΔQ small



ΔP small, ΔQ large

General rule: A demand shock is *passed through* to the less elastic curve: if supply is rigid, it all goes into the price; if it is elastic, it all goes into quantities.

Example: Semiconductors (2020–22) (1)

A documented case of a **simultaneous shock** in opposite directions.

Demand shock ($D \uparrow$):

- Remote working during the pandemic: demand for PCs, laptops, webcams.
- Gaming boom with new consoles (PS5, Xbox Series X/S, November 2020).
- Transition to electric cars, much more chip-intensive than traditional ones.

Supply shock ($S \downarrow$):

- Lockdowns in Taiwan and Southeast Asia, where production is concentrated (TSMC, UMC).
- Taiwan's 2021 drought: chip production requires enormous quantities of water.
- Fire at Renesas in Japan (March 2021), a major automotive supplier.

Example: Semiconductors (2020–22) (2)

Observed effects:

- Chip spot prices: +200% in some categories.
- Supply *lead times*: from about 12 to about 52 weeks.
- Global car production: a loss of about 7.7 million vehicles in 2021 (AlixPartners).

Reading it through the model

$D \uparrow$ and $S \downarrow$ imply $\uparrow P^*$ with certainty, ΔQ^* ambiguous. Empirically: Q fell for advanced chips (*leading edge*, automotive, servers) because the supply shock dominated; it rose for standard chips thanks to new fabs.

Profit as a Market Signal

In the short run firms can make positive profits or losses. But in the long run, with free entry and exit, a powerful mechanism is triggered.

Profit as a signal

- $\pi > 0$: the industry earns more than average $\Rightarrow$ new firms **enter**
- $\pi < 0$: the industry earns less than average $\Rightarrow$ existing firms **exit**
- $\pi = 0$: normal return $\Rightarrow$ no incentive to enter or exit

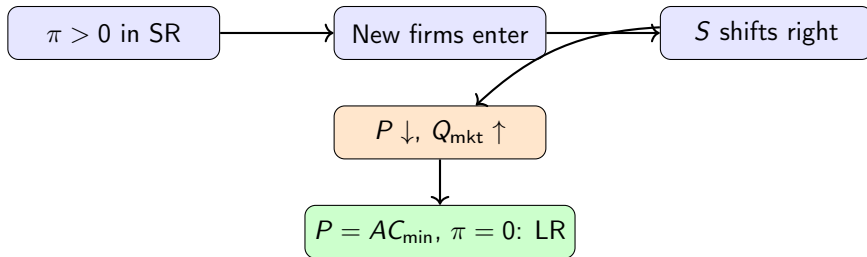
LR assumptions with free entry:

- Firms can freely enter and exit (no barriers)
- Every new firm has the same cost function as existing ones
- The process ends when $\pi = 0$ for all active firms

The LR price tends to the minimum of average cost: $P_{LR}^* = AC_{\min}$. Firms make zero economic profit, but earn the normal return on capital.

Mechanism with $\pi > 0$: Entry

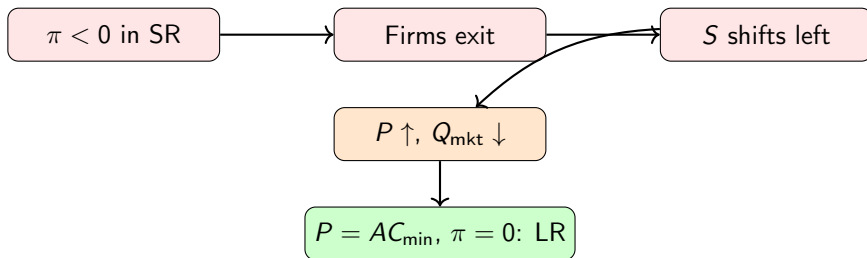
Scenario: $P_{SR} > AC_{min}$, firms make $\pi > 0$.



Important note: in the LR each firm keeps producing q^* (the quantity at the minimum of AC). The increase in market quantity comes from the increase in the **number of firms**, not from the size of each.

Mechanism with $\pi < 0$: Exit

Scenario: $P_{SR} < AC_{min}$, firms make $\pi < 0$.



LR equilibrium condition

In long-run equilibrium with free entry:

$$P_{LR}^* = AC_{min} \Rightarrow \pi = 0$$

Each active firm produces q^* such that $P_{LR}^* = MC(q^*) = AC_{min}$.

LR Equilibrium: Computation

With $C(q) = q^2 + 4q + 9$

Step 1. Find $AC_{\min}$ from the condition $MC = AC$:

$$2q + 4 = q + 4 + \frac{9}{q} \Rightarrow q = \frac{9}{q} \Rightarrow q^2 = 9 \Rightarrow q^* = 3$$

$$P_{LR}^* = AC(3) = 3 + 4 + 3 = 10$$

Step 2. With $P_{LR}^* = 10$ and demand $Q^D = 120 - 4P$:

$$Q_{\text{mkt}}^* = 120 - 4 \cdot 10 = 80$$

Step 3. Number of firms:

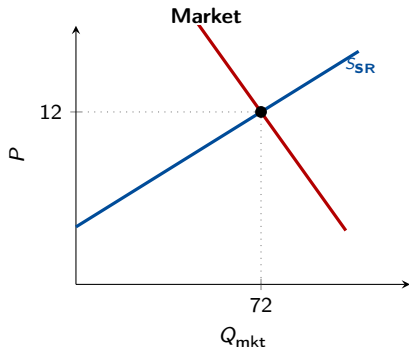
$$n^* = \frac{Q_{\text{mkt}}^*}{q^*} = \frac{80}{3} \approx 26.7 \Rightarrow n^* = 26 \text{ or } 27 \text{ firms}$$

Zero-profit check

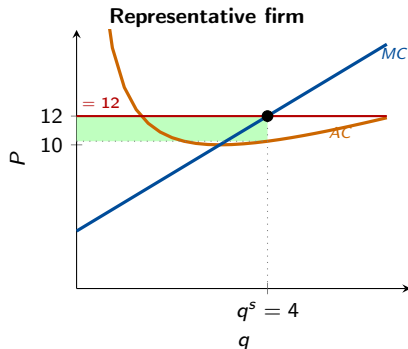
$$\pi = P_{LR}^* \cdot q^* - C(q^*) = 10 \cdot 3 - (9 + 12 + 9) = 30 - 30 = 0 \quad \checkmark$$

Market and Representative Firm: SR

Two graphs **read together**: the market determines P^* ; the firm takes P^* as given. With $n = 18$ firms in the SR and $C(q) = q^2 + 4q + 9$.



$$P_{\text{SR}}^* = 12, Q^* = 72$$

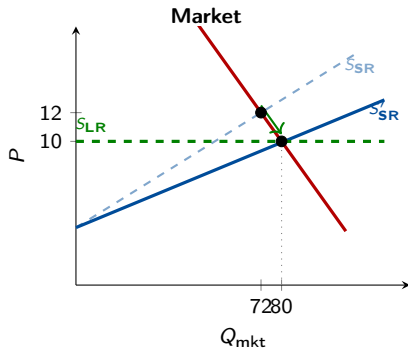


$$q^s = 4, AC(4) = 10.25, \pi = 7$$

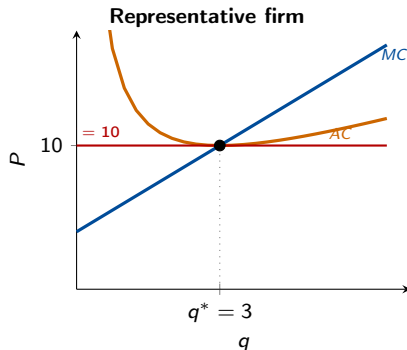
Reading. At $P = 12$ the firm produces where $MC = P$, obtaining $q^s = 4$. Average cost at that quantity is $AC(4) = 10.25 < 12$: profit per firm $= (12 - 10.25) \cdot 4 = 7$. With 18 firms, market quantity is $18 \cdot 4 = 72$, which coincides with $Q^D(12) = 120 - 48 = 72$. **The market is in SR equilibrium, but firms make $\pi > 0$.**

Market and Representative Firm: LR

In the LR new firms enter until $\pi = 0$. Market supply shifts right (from $n = 18$ to $n \approx 27$) and the price returns to $AC_{\min} = 10$.



$$P_{\text{LR}}^* = 10, Q^* = 80, n = 27$$



$$q^* = 3, AC(3) = 10, \pi = 0$$

Reading. The point $(q^*, P_{\text{LR}}^*) = (3, 10)$ is the *minimum* of the AC curve: in the LR each firm is *squeezed* to its efficient scale. LR supply is horizontal at $P = 10$ (constant-cost industry): the market responds to demand shocks by changing n , not P .

Short, Medium, Long Run: What Adjusts?

The concept of *run* in economics is not chronological but **analytical**: it indicates which variables are free to adjust.

Run	What is fixed	What adjusts
Market	n , K , output already produced	Price P only
Short (SR)	n , K , technology	P and q per firm (variable inputs)
Medium	n , technology	P , q , K
Long (LR)	Nothing	P , q , K , n , technology

Concrete examples of different LR horizons:

- *Fresh fish, fruit, cut flowers*: market period dominant (perishability $\Rightarrow$ quantity fixed on the day, price adjusts within hours).
- *Software, digital services*: LR within a few months (low barriers to entry, redeployable human capital).
- *Manufacturing*: LR typically 1–3 years (time to build and test plants).
- *Quality wine, cargo ships, nuclear power plants*: LR 5–15 years (mature vineyards, hulls, licenses).

The model we will use

In the rest of the course we consider only two poles: SR (number of firms n and capital fixed) and LR (free entry, $\pi = 0$)

Demand Shock in the LR: Step by Step

Scenario: demand increases from $Q^D = 120 - 4P$ to $Q^{D'} = 140 - 4P$.

Short run (fixed number of firms, $n = 27$, each firm produces $q^s(P) = (P - 4)/2$):

$$Q_{SR}^S = 27 \cdot \frac{P - 4}{2} = 13.5P - 54$$

$$140 - 4P = 13.5P - 54 \Rightarrow 194 = 17.5P \Rightarrow P_{SR}^{**} \approx 11.09$$

Firms make $\pi > 0$ ($P > AC_{\min} = 10$) $\Rightarrow$ new firms enter.

Long run (free entry, $\pi = 0$):

$$P_{LR}^* = AC_{\min} = 10 \quad (\text{unchanged!})$$

$$Q_{LR}^{* \text{ new}} = 140 - 4 \cdot 10 = 100$$

$$n_{\text{new}}^* = \frac{100}{3} \approx 33 \text{ firms} \quad (\text{there were 27})$$

Key result

In the LR the price **returns** to the break-even level. The permanent effect of a demand shock is on total quantity and on the number of firms, **not** on the price.

Constant-Cost Industry: Formal Derivation of the LR Curve

Why does the LR supply curve have that shape? It depends on how input costs vary with the size of the industry.

Notation: let $w(n)$ be the equilibrium wage of the variable input when there are n firms in the industry. The common technology yields $AC_{\min}(w)$.

Constant-cost industry: $\partial w / \partial n = 0 \Rightarrow AC_{\min}$ unchanged as n grows.

$$P_{LR}^* = AC_{\min}(w_0) \quad \forall n \quad \Rightarrow \text{horizontal LR supply}$$

Increasing-cost industry: $\partial w / \partial n > 0$. As n grows, wages rise:

$$P_{LR}^*(n) = AC_{\min}(w(n)), \quad \frac{dP_{LR}^*}{dn} = \frac{\partial AC_{\min}}{\partial w} \cdot \frac{\partial w}{\partial n} > 0$$

The LR curve is increasing: each new LR equilibrium has a higher P than the previous one.

Decreasing-cost industry: $\partial w / \partial n < 0$ (external economies). P_{LR}^* decreases. Decreasing LR curve. Empirically rare.

Implication

The shape of the LR supply curve is a structural property of the input market, not of the technology of individual firms.

Constant-Cost Industry

Assumption

The expansion of the industry does not change input prices $\Rightarrow AC_{\min}$ stays constant.

Mechanism: shock $D \uparrow \Rightarrow$ SR: $P \uparrow, \pi > 0 \Rightarrow$ firms enter $\Rightarrow P$ returns to the unchanged $AC_{\min}$.

LR supply curve: horizontal at $P = AC_{\min}$. $\varepsilon_S^{\text{LR}} = \infty$.

Typical examples. Manufacturing industries using widely available inputs (standard plastics, commodity electronic components). The expansion of one sector does not alter the world price of steel or oil.

Implication for consumers: in the long run, an increase in demand translates *entirely* into higher quantity, *not* into a higher price. The market adjusts through quantity.

Increasing-Cost Industry

Assumption

The expansion of the industry increases the demand for specialized inputs $\Rightarrow w, r$ rise $\Rightarrow AC_{\min}$ increases.

Mechanism: shock $D \uparrow \Rightarrow$ firms enter $\Rightarrow$ competition for scarce inputs $\Rightarrow w, r$ rise $\Rightarrow$ new higher $AC_{\min} \Rightarrow$ new $P_{LR}^* >$ old.

LR supply curve: increasing (but flatter than SR supply). $0 < \varepsilon_S^{LR} < \infty$.

Typical examples:

- Oil industry: expansion requires ever deeper and more costly wells.
- Quality wine industry: expansion requires premium vineyard land, whose price rises.
- Skill-intensive sectors: expansion requires more specialists, whose wages rise.

Decreasing-Cost Industry

Assumption

The expansion of the industry generates external economies (infrastructure, technological spillovers, industrial clusters) $\Rightarrow AC_{\min}$ falls.

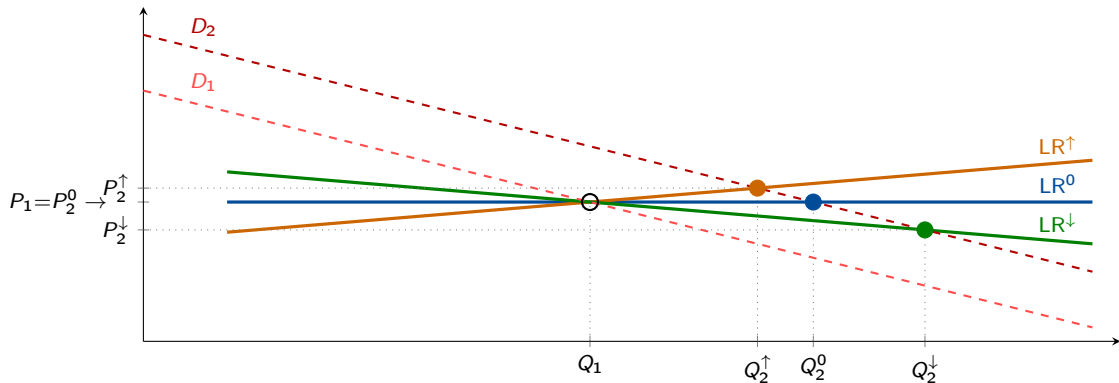
Mechanism: shock $D \uparrow \Rightarrow$ firms enter $\Rightarrow$ external economies develop $\Rightarrow AC_{\min}$ decreases $\Rightarrow$ new $P_{LR}^* < \text{old}$.

LR supply curve: decreasing (not to be confused with the demand curve!). Rare.

Typical examples:

- Semiconductors (Moore's Law): industry growth has reduced chip production costs decade after decade.
- Renewable energy: scale has reduced the cost of solar panels by 90% in 15 years.
- Software: network economies and economies of scale reduce average cost.

Comparing the Three Types: Visual Summary



- **Constant** (LR^0): $\Delta P = 0$; the entire adjustment runs through quantity via entry of new firms at unchanged $AC_{\min}$. $\varepsilon_S^{LR} = \infty$.
- **Increasing** ($LR^\uparrow$): $\Delta P > 0$, ΔQ *dampened*; the rise in input prices redistributes surplus from consumers to the owners of scarce inputs (inframarginal rents). $0 < \varepsilon_S^{LR} < \infty$.
- **Decreasing** ($LR^\downarrow$): $\Delta P < 0$, ΔQ *amplified*; external economies lower $AC_{\min}$ and benefit consumers through both price and quantity.

Example: Free Entry in the Restaurant Industry

Urban restaurants are a textbook case of **perfect competition with free entry** in an *increasing-cost* industry.

Features of the sector:

- Many independent firms (low market concentration).
- Modest barriers to entry: initial capital of 50–150k euros, licenses obtainable.
- Differentiated but highly substitutable product.

Stylized facts (average Italian metropolitan area):

- Annual opening rate: about 15% of existing establishments.
- Annual closure rate: about 13%.
- About 60% of new businesses close within the first 3–5 years.

Why an increasing-cost industry. When local demand grows (tourism, gentrification), new restaurants enter. Demand for well-located commercial space rises, rents increase, $AC_{\min}$ rises. The result: industry expansion structurally raises the equilibrium price of meals.

The mechanism at work

Central Florence or Venice: a tourism boom leads to an increase in demand, so new restaurants enter; competition for central premises drives up rents, and meal prices become structurally higher than in less touristy cities. The LR equilibrium is restored at a *new*, higher P_{LR}^* .

Exercise 1—SR Equilibrium

Market with $Q^D = 60 - 4P$ and $Q^S = 16P$.

- (a) Find P^* and Q^* .
- (b) At $P = 1$: compute excess demand/supply. Does the price rise or fall?
- (c) At $P = 5$: compute excess demand/supply.
- (d) Demand increases to $Q^{D'} = 80 - 4P$. Find the new equilibrium.

Exercise 1—Solution

(a) $60 - 4P = 16P \Rightarrow 60 = 20P \Rightarrow P^* = 3, Q^* = 48$

(b) $P = 1$: $Q^D = 56, Q^S = 16$. Excess demand $= 56 - 16 = 40$.
The price **rises** toward 3.

(c) $P = 5$: $Q^D = 40, Q^S = 80$. Excess supply $= 80 - 40 = 40$.
The price **falls** toward 3.

(d) $Q^{D'} = 80 - 4P = 16P \Rightarrow 80 = 20P \Rightarrow P^{**} = 4, Q^{**} = 64$

Demand increases $\Rightarrow P$ and Q both increase ✓

Exercise 2—LR Equilibrium

All firms in the industry have $C(q) = q^2 + 6q + 25$. Demand: $Q^D = 120 - 2P$.

- (a) Find P_{LR}^* and q^* (optimal quantity per firm).
- (b) Find Q_{mkt}^* and the number of firms n^* .
- (c) Demand increases to $Q^{D'} = 160 - 2P$. Find the new LR.
- (d) How many firms have entered?

Exercise 2—Solution (a) and (b)

(a) $AC_{\min}$: condition $MC = AC$:

$$2q + 6 = q + 6 + \frac{25}{q} \Rightarrow q = \frac{25}{q} \Rightarrow q^2 = 25 \Rightarrow q^* = 5$$

$$P_{LR}^* = AC(5) = 5 + 6 + 5 = 16$$

(b) $Q_{\text{mkt}}^* = Q^D(16) = 120 - 32 = 88$.

$$n^* = \frac{88}{5} = 17.6 \text{ (say 18 firms)}$$

Check: $\pi = 16 \cdot 5 - (25 + 30 + 25) = 80 - 80 = 0 \checkmark$

Exercise 2—Solution (c) and (d)

(c) With $Q^{D'} = 160 - 2P$, in the LR (constant-cost industry):

$$P_{LR}^* = 16 \quad (\text{unchanged: same technology, same costs})$$

$$Q_{\text{mkt}}^{*\text{new}} = 160 - 32 = 128$$

$$n_{\text{new}}^* = \frac{128}{5} = 25.6 \approx 26 \text{ firms}$$

(d) Firms entered: $26 - 18 = 8$ firms (approximately).

Summary of the shock

	P^*	Q_{mkt}^*	n^*
Before the shock	16	88	18
After the shock (LR)	16	128	26

The price does not change in the LR. Only the size of the industry changes.

Exercise 3—Qualitative Analysis

For each shock: indicate (i) which curve shifts and in which direction, (ii) the effect on P^* and Q^* in the SR, (iii) what happens in the LR.

- (a) The government introduces a tax of €3 per unit produced.
- (b) A new technology reduces variable production costs by 20%.
- (c) Consumers' income falls (normal good).
- (d) The price of a key input doubles.

- (a)** Tax $\Rightarrow MC$ rises $\Rightarrow S$ shifts left. SR: $P \uparrow$, $Q \downarrow$. LR: $\pi < 0$, some firms exit; new higher P_{LR} .
- (b)** Technology $\Rightarrow MC$ falls $\Rightarrow S$ shifts right. SR: $P \downarrow$, $Q \uparrow$. LR: $\pi > 0$, new firms enter; P_{LR} falls to the new lower AC_{min} .
- (c)** Income $\downarrow \Rightarrow D$ shifts left. SR: $P \downarrow$, $Q \downarrow$. LR (constant costs): P returns to AC_{min} , firms exit.
- (d)** Input $\uparrow \Rightarrow MC$ and AC rise $\Rightarrow S$ shifts left. SR: $P \uparrow$, $Q \downarrow$. LR: firms exit; new higher P_{LR} .

Exercise 4—Surplus and Comparative Statics

Coffee market: $Q^D = 180 - 6P$ (inverse: $P_D = 30 - Q/6$) and $Q^S = 12P$ (inverse: $P_S = Q/12$).

- (a) Find the equilibrium (P^*, Q^*) and compute CS, PS, TS.
- (b) Draw the equilibrium and the CS and PS areas.
- (c) An increase in income raises demand to $Q^{D'} = 216 - 6P$. Find the new equilibrium and recompute CS, PS, TS.
- (d) Compute ΔCS , ΔPS , ΔTS . Who gets the larger share of the gain? Explain in terms of relative elasticity.

Exercise 4—Solution

(a) $180 - 6P = 12P \Rightarrow 180 = 18P \Rightarrow \boxed{P^* = 10, Q^* = 120}$. Demand intercept: $P_{\max}^D = 30$.

$$CS = \frac{1}{2} \cdot 120 \cdot (30 - 10) = 1200; \quad PS = \frac{1}{2} \cdot 120 \cdot 10 = 600; \quad TS = 1800$$

(b) Graph: triangles CS (above $P^* = 10$, below D , from $Q = 0$ to $Q = 120$) and PS (below $P^* = 10$, above S , from $Q = 0$ to $Q = 120$).

(c) $216 - 6P = 12P \Rightarrow 216 = 18P \Rightarrow \boxed{P^{**} = 12, Q^{**} = 144}$. New intercept: $P_{\max}^{D'} = 36$.

$$CS' = \frac{1}{2} \cdot 144 \cdot (36 - 12) = 1728; \quad PS' = \frac{1}{2} \cdot 144 \cdot 12 = 864; \quad TS' = 2592$$

(d) $\Delta CS = +528$, $\Delta PS = +264$, $\Delta TS = +792$. The consumers' gain is **exactly twice** that of producers.

Interpretation

Slope of inverse supply = $1/12$; slope of inverse demand = $1/6$. Supply is **twice as flat** as demand, so it responds to the shock mainly through quantity: $\Delta Q = 24$ units versus $\Delta P = 2$ euros. The welfare gain is distributed in direct proportion to the slopes (i.e., inversely to the elasticities), hence 2:1 in favor of consumers.

General Summary

SR equilibrium

- $Q^D(P^*) = Q^S(P^*)$; found by solving the system
- Disequilibrium: excess demand $\Rightarrow P \uparrow$; excess supply $\Rightarrow P \downarrow$

Surplus and welfare

- CS = area between D and P^* ; PS = area between P^* and S ; TS = CS+PS
- First Welfare Theorem: the competitive equilibrium maximizes TS

Comparative statics

- $D \uparrow \Rightarrow P \uparrow, Q \uparrow$; $S \uparrow \Rightarrow P \downarrow, Q \uparrow$
- The incidence of a shock depends on relative elasticities

LR adjustment

- $\pi > 0 \Rightarrow$ entry; $\pi < 0 \Rightarrow$ exit; LR: $P^* = AC_{\min}$, $\pi = 0$
- Demand shock in the LR (constant costs): P^* unchanged; Q^* and n^* change

LR supply curve

- Constant costs (horizontal), increasing (upward-sloping), decreasing (downward-sloping)

Skills Acquired

- ✓ Define competitive equilibrium and the Walrasian mechanism
- ✓ Compute (P^*, Q^*) by solving $Q^D = Q^S$
- ✓ Interpret excess demand/supply and apply comparative statics
- ✓ Compute CS, PS, TS and recognize the efficiency of equilibrium
- ✓ Describe the LR adjustment mechanism with *free entry*
- ✓ Determine $P_{LR}^* = AC_{\min}$, Q^* and n^* in the LR equilibrium
- ✓ Classify constant-, increasing-, and decreasing-cost industries

Optional Homework Exercises

1. $Q^D = 200 - 4P$, $Q^S = 6P$. Find P^* , Q^* , CS, PS, TS. If D falls to $Q^D = 160 - 4P$, find the new equilibrium and the welfare changes.
2. $C(q) = 2q^2 + 8q + 18$. Find P_{LR}^* and q^* . With $Q^D = 500 - 10P$, find n^* .
3. In a constant-cost market, demand increases by 30%. Describe the SR and LR adjustment. What happens to P^* , Q^* , n^* , CS, PS?
4. Explain why under perfect competition with free entry LR economic profit is zero, yet firms keep operating.
5. $C(q) = q^2 + 6q + 25$. The SR market price is $P = 20$. (a) Find q^* and π per firm. (b) Is the market in LR equilibrium? (c) What P is consistent with the LR?
6. Market with $Q^D = 300 - 10P$, $Q^S = 20P$. Compare two scenarios: inelastic supply ($Q^S = 100$, constant) vs elastic supply ($Q^S = 40P$). In which case does a positive demand shock raise the price more? Explain.